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INTERVENTION, EVALUATION, AND POLICY STUDIES

Investigating the Efficacy of a Core Kindergarten Mathematics Curriculum to Improve Student Mathematics Learning Outcomes

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Abstract: This study examined the efficacy of a core kindergarten mathematics program, Early Learning in Mathematics (ELM), a 120-lesson program with four content strands: (a) number operations, (b) geometry, (c) measurement, and (d) vocabulary. The study utilized a randomized block design, with 129 classrooms randomly assigned within schools to treatment or control conditions. Measures of achievement were collected in the fall (pretest) and spring (posttest) in kindergarten and in the winter of first grade (follow-up). Although students in ELM classrooms did not differ significantly from students in control classrooms, we expected and found that the effects of ELM depended on students' initial achievement. Initial achievement moderated condition effects for Test of Early Mathematics Ability (TEMA) scores ($p = .0039$), but not the Early Numeracy–Curriculum-Based Measurement ($p = .1887$). We found no effects on follow-up first-grade scores from the Stanford Achievement Test Tenth Edition Problem Solving subtest ($p = .9737$) or Procedures subtest ($p = .6336$).

Keywords: Mathematics, number sense, prevention, learning disabilities

An increased focus has been paid to improving the mathematics instruction provided to students in the United States. In large part, this focus has been driven by concern over persistent and troubling low performance on both national and international assessments of mathematics achievement (National Mathematics Advisory Panel [NMAP], 2008; National Research Council, 2001; Schmidt, Houang, & Cogan, 2002). Results from the 2013

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National Assessment for Educational Progress (NAEP) indicate that only 42% of fourth graders were deemed at or above proficient in mathematics and 17% were below basic (National Center for Education Statistics, 2013). Difficulties in mathematics achievement are particularly severe for students from low income and minority backgrounds and those with learning disabilities. Results from international studies comparing American students to their counterparts across the world show American students' performance falling in the middle to lower portion of the achievement spectrum (Trends in International Mathematics and Science Study [TIMSS]; Olson, Martin & Mullis, 2008). In part these findings are disconcerting because a vast range of opportunities within multiple settings are dependent upon a fundamental understanding of mathematics from within school settings (e.g., advanced science classes) to growing employment opportunities in fields heavily dependent upon mathematics (National Science Board, 2008). The purpose of this study was to examine the impact of a core kindergarten program, Early Learning in Mathematics (ELM), on student mathematics achievement with a specific focus on meeting the learning needs of at-risk students.

THE IMPORTANCE OF KINDERGARTEN

Longitudinal research suggests that students who perform poorly in mathematics at the end of kindergarten are likely to continue to struggle throughout elementary school (Bodovski & Farkas, 2007; Duncan et al., 2007; Hanich, Jordan, Kaplan, & Dick, 2001; Morgan, Farkas, & Wu, 2009). Using a nationally representative sample of students from the Early Childhood Longitudinal Study–Kindergarten Cohort (ECLS-K), Morgan et al. (2009) found that students who were below the 10th percentile at entrance and exit from kindergarten (considered an indicator of a learning disability in mathematics—MLD) had a 70% chance of remaining below the 10th percentile five years later. Their overall mathematics achievement in fifth grade remained two standard deviation units below students who did not demonstrate an MLD profile in kindergarten. Unless the needs of at-risk students are addressed in kindergarten, they are likely to persist and become more difficult to remediate over time as students exit kindergarten without a solid foundation upon which to build an increasingly complex understanding of mathematics (Geary, 1993; Jordan, Kaplan, & Hanich, 2002; Lyon et al., 2001; Morgan et al., 2009). However, if mathematics trajectories can be altered in kindergarten the lasting impact is potentially profound and could fundamentally alter long-term mathematics outcomes (Morgan et al., 2009).

MATHEMATICS INSTRUCTION IN KINDERGARTEN

There have been efforts to improve mathematics instruction and impact student mathematics outcomes in kindergarten. Much of the work in the area has focused on the development of number sense. Number sense is a construct (Dehaene, 1997) that refers broadly to a child's fluidity and flexibility with using and manipulating numbers, an almost intuitive sense of what numbers mean, and the ability to use that sense to make mathematical comparisons (Berch, 2005; Gersten & Chard, 1999). There is some support that instruction focused on building number sense leads to significant reductions in mathematics failure in the primary grades. Seminal work by Griffin and colleagues (Griffin, 2004; Griffin, Case, & Siegler, 1994) to develop and evaluate a program constructed around the development of number sense indicated early promise in building a deep understanding of number and a reduction

in mathematics failure. Although not rigorously evaluated and not clearly specified in terms of instructional delivery formats, such early efforts laid the groundwork for a focus on the development of number sense as a critical instructional variable in kindergarten. Specifically because many at-risk students lack an understanding of number, interventions have tended to focus on building that understanding through the delivery of additional content in small group formats (e.g., Baroody, Eiland, & Thompson, 2009; Bryant et al., 2011; Clarke et al., 2011; Dyson, Jordan, & Glutting, 2011). For example, Dyson and colleagues (2011) designed a 24-lesson small group intervention for low-income kindergarten students focused on the development of number sense with a specific focus on whole number concepts. Results from their work indicate promise, with intervention students showing greater gains than control students on posttest and delayed posttest measures. But as Dyson et al. (2011) noted, the number of intervention studies targeting kindergarten at-risk students in mathematics are few in overall number and the number of studies that utilize designs needed to draw causal conclusions regarding program efficacy are limited and remain the exception rather than the rule (Dyson et al., 2011; Gersten, Chard, et al., 2009; NMAP, 2008; What Works Clearinghouse, 2011).

Although there is a growing but limited body of research on early elementary mathematics interventions (Gersten, Beckmann, et al., 2009), less research has been conducted on core curricula effectiveness. In a review of the literature, we found no rigorously designed research studies examining the impact of a core kindergarten mathematics curriculum. Examining core curricula effectiveness is critical because in kindergarten it is likely that the core curricula will represent the primary exposure to mathematics for students, including those at risk.

THEORY OF CHANGE MODELS IN CURRICULUM DEVELOPMENT AND EVALUATION

Critical to the development of research-based curricula are frameworks that link curriculum development efforts and subsequent evaluations to an underlying theory of change (Clements, 2007) rooted in relevant theories of learning and the current research base (e.g., Donovan & Bransford, 2005). A strong theory of change also addresses the variables hypothesized to mediate and moderate the impact of an intervention. Mediating variables represent the processes that comprise an intervention. Moderating variables are student and teacher factors that may potentially change the relationship between an intervention and student outcomes. A theory of change that articulates mediators and moderators offers researchers the opportunity to unpack the “black box” of instruction (MacKinnon & Luecken, 2008) and examine the conditions under which an intervention may be more effective, including with subsets of students such as those at risk in mathematics.

ELM OVERVIEW AND THEORY OF CHANGE

Our efforts to develop and evaluate ELM were guided by an underlying theory of change model. As depicted in Figure 1, the theory of change for the ELM intervention is comprised of three key levels: (a) internal dimensions including curriculum content and explicit and systematic instructional design principles, (b) an external dimension designated as quantity and quality of teacher-student interactions, and (c) proximal and distal student outcomes of conceptual understanding, procedural fluency, and general mathematics achievement. The

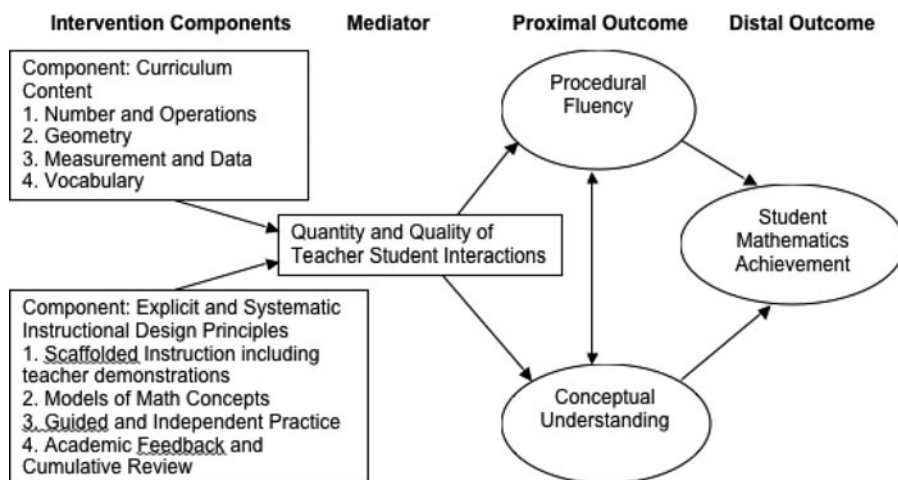


Figure 1. Early learning in mathematics theory of change.

unique features of the curriculum are contained in the two internal dimensions and one external dimension. Internal dimensions are *curriculum content* and *curriculum design*. Both have been constructed to maximize opportunities for student learning.

ELM Curriculum Content

The ELM curriculum is a core mathematics program designed to support the range of learners in kindergarten classrooms. ELM's first internal dimension, curriculum content, is intended to provide students with a deep understanding of essential mathematical concepts and principles. Based on the recommendation of National Council of Teachers of Mathematics (2006), the NMAP (2008), and the Common Core State Standards Initiative (CCSSI, 2010) rather than attempt to teach a comprehensive list of all mathematics to be learned in school, ELM provides a narrower but more focused coverage of the critical conceptual knowledge and skills all students should master by the end of each grade level. The program includes 120 45-minute lessons with supplemental daily 15-minute calendar activities. A typical ELM lesson includes four to five activities, each of which focuses on one of three content strands: number and operations, measurement and data, and geometry. Across the three content strands, a fourth strand, vocabulary, is integrated to increase the amount of student mathematics discourse and use of mathematics-specific vocabulary. ELM is fully aligned with the learning objectives for kindergarten as specified in the CCSSI (2010). The first activity introduces or reviews a math concept or skill that is central to the lesson's overall objective. The second and third activities in ELM often focus on extending a student's understanding and work with the content introduced in the first activity. The fourth activity in a typical lesson involves active review of previously learned material from a different strand. The last activity for the day involves a paper pencil review led by the teacher and includes a take-home note in either English or Spanish. Every fifth lesson in ELM is focused on problem solving and engages students in utilizing known math content in an applied setting. As specified in the CCSSI (2010) and recommended by experts (NMAP, 2008),

although ELM covers multiple content areas, the strongest focus is on the development of whole number concepts and understanding. Support for the curriculum content internal dimension of ELM's content focus is based on expert opinion because there is limited research on core curriculum (Gersten, Beckmann, et al., 2009). Thus guidance is largely provided by a number of seminal documents released, including focal points for instruction by the leading organization for mathematics instruction (National Council of Teachers of Mathematics, 2006), nationally commissioned reports (NMAP, 2008) and most recently attempts to establish national curriculum standards (CCSSI, 2010).

ELM Curriculum Design

The second internal dimension of ELM, explicit and systematic instructional design principles, is empirically supported by a growing body of research and has begun to set forth the types of instructional strategies or instructional design features that are most effective for students who struggle with mathematics or have a mathematics learning disability. Three recent meta-analyses on low-achieving students (Baker, Gersten, & Lee, 2002) and students with learning disabilities (Gersten, Chard, et al., 2009; Kroesbergen & Van Luit, 2003) identified 15, 58, and 38 studies, respectively. Summaries of these findings (Gersten, Chard, et al., 2009; NMAP, 2008) have attempted to articulate what instruction that incorporates these elements would look like in classrooms. Foremost, instruction would be explicit and systematic. Explicit and systematic instruction builds proficiency in a scaffolded, logical, and coherent manner, initially demonstrating critical concepts to students and then providing multiple opportunities to work with that concept to mastery. Within ELM, teachers provide models or think-alouds (Gersten, Beckmann, et al., 2009) as they solve problems in which concepts and processes are made overt. Following teacher models, students solve similar problems and are provided with specific and immediate feedback by the teacher as they verbalize and explain their solutions and understanding of the underlying mathematical concepts. Finally, ELM includes frequent and cumulative review, including the use of distributed practice (Pashler et al., 2007) of key concepts to ensure retention and generalization of acquired mathematics knowledge.

Teacher-Student Interactions and Student Learning Outcomes

The external dimension of ELM is how the curriculum is actually taught by classroom teachers on a regular day-to-day basis to the full range of students in general education classrooms (quantity and quality of teacher-student interactions). That is, we expect classroom instruction variables will mediate student learning, and that the content and design of ELM will directly facilitate frequent opportunities for teachers to engage in high-quality instructional interactions with students. Together, these dimensions of ELM—content, design, and instructional interactions—enable students to develop a conceptual understanding of and procedural fluency with critical mathematics. Conceptual knowledge is an understanding of the relationship between representations of math concepts and abstract symbols, while procedural fluency entails automaticity of math procedures (National Research Council, 2001; Wu, 1999). Our theory of change views these knowledge forms as interdependent constructs that develop in tandem (Rittle-Johnson, Siegler, & Alibali, 2001; Wu, 1999). Reciprocal development occurs because as students increase their conceptual understanding of math concepts (e.g., place value), their flu-

ency in solving math problems (e.g., addition with renaming) increases in a corresponding fashion. Development of skills and strategies in these two knowledge forms contributes directly to student performance on proximal and distal measures of mathematics achievement.

PURPOSE OF THE STUDY

This article examines the effects of a kindergarten core curriculum, Early Learning in Mathematics (ELM), on gains in student achievement from the fall of kindergarten (T_1 , pretest) to the spring of kindergarten (T_2 , immediate posttest) and to the winter of first grade (T_3 , follow-up), roughly eight months after the end of ELM instruction. The study will help address the gap in research on core curricula overall and specifically the lack of research on core curricula in kindergarten. We investigate three a priori research questions:

1. Does the ELM curriculum produces greater achievement gains at the end of kindergarten and at the end of Grade 1 than standard district mathematics instruction?
2. Does the impact of the ELM curriculum on student achievement depend on the state (i.e., suburban schools in Oregon vs. urban schools in Texas)?
3. Does the impact of the ELM curriculum on student achievement depend on their initial scores at the beginning of kindergarten?

We hypothesize that ELM will produce greater achievement gains at the end of kindergarten and Grade 1 than standard district mathematics instruction. Because ELM was designed to fit into a wide variety of educational contexts and meet the needs of all learners, we expect the direction of the effect to be the same for both states. Lastly, we hypothesize that at-risk students in ELM classrooms will benefit to a greater degree than students not at risk due to the design of the curriculum.

METHOD

Design

We examined the efficacy of the ELM curriculum by implementing a randomized control trial in 129 classrooms across two states, Oregon and Texas. The Oregon sample participated in 2008–2009 and Texas participated in 2009–2010. Participating kindergarten classrooms were randomly assigned to treatment or control conditions, blocking on school and whether kindergarten was full or half day. In treatment classrooms (68 classrooms), classroom teachers provided ELM instruction. Teachers in control classrooms (61 classrooms) utilized their existing math curriculum. Student-level math achievement data was collected during the fall (T_1 , pretest) and spring (T_2 , posttest) of the kindergarten implementation year and at roughly six months into the students' first-grade year (T_3 , follow-up). Data from classroom observations and teacher self-report were also analyzed. Our primary analysis framework is a group-randomized trial (Murray, 1998, 2001) with students nested within classrooms and classrooms nested within condition.

Table 1. Descriptive information for classrooms and teachers by region and condition

	ELM		Control		Total
	Oregon	Texas	Oregon	Texas	
Number of classrooms	34	34	30	31	129
Teacher gender					
Female	32	33	31	31	127
Male	2	1	0	0	3
Teacher ethnicity					
White	33	17	26	16	92
Hispanic	0	11	2	9	22
African American	0	5	0	6	11
Native American	0	0	1	0	1
Asian American	0	1	0	0	1
Teacher education					
Master's degree	22	6	14	7	49
Special education	2	3	0	1	6
Completed 3 or more college math courses	12	6	7	5	30
Completed college algebra	17	24	13	14	68
Teacher experience (have taught kindergarten for 4+ years)	16	22	18	17	73
Number of students in class <i>M</i> (SD)	22.4 (3.9)	20.2 (3.2)	21.7 (3.4)	18.8 (3.4)	20.8 (3.7)
School type					
Public	34	17	30	16	97
Private	0	11	0	9	20
Charter	0	6	0	6	12
Program structure					
Full-day program	26	34	21	31	112
Half-day program	8	0	9	0	17
Minutes on math instruction					
21 to 40 minutes per day	8		16		
41 to 60 minutes per day	19	10	9	7	
61+ minutes per day		18		19	

Note. The 129 participating classrooms were taught by 130 teachers in 46 schools. Two teachers taught one class, each working a half-day schedule. The school type and program structure represent the number of classrooms and not the number of schools.

Participants

Participating kindergarten classrooms ($n = 129$) were drawn from 46 urban, suburban, and charter schools in Oregon and Texas (see Table 1). The majority of schools offered 41–60 minutes of kindergarten math instruction per day (see Table 1). Participating schools received a stipend.

The study sample included 2,598 kindergarten students (1,401 treatment, 1,197 control) and their classroom teachers ($n = 130$). As shown in Table 2, students in both conditions were similar in terms of age. Teachers implementing ELM ($n = 68$) received professional

Table 2. Descriptive statistics for students by condition

Measure	ELM		Control	
	Oregon	Texas	Oregon	Texas
Age at T ₁ (Months) <i>M</i>	65.5	67.7	66.5	68.0
(<i>SD</i>)	(3.77)	(4.26)	(3.77)	(4.22)
Sample Size	730	671	619	578

development focused on implementing the ELM curriculum, ELM materials, a stipend, and payment for training hours conducted outside their regular workday. Teachers in control classrooms ($n = 61$) taught mathematics as they would typically. They received a stipend and, at the conclusion of the study, were given access to the ELM curriculum. Table 1 depicts self-reported teacher demographics.

Measures

Student Measures

The Early Numeracy–Curriculum-Based Measurements (EN-CBM) measures (Clarke & Shinn, 2004) and the Test of Early Mathematics Ability, Third Edition (TEMA-3; Pro Ed, 2007) were individually administered at both pretest and posttest to all students. The EN-CBM measures were used as a proximal measure of procedural fluency and the TEMA was used as a proximal measure of conceptual understanding. During the winter of first grade (follow-up), all students took the *math problem solving* and *math procedures* subtests of the Stanford Achievement Test–Tenth Edition (SAT-10; Harcourt Brace Educational Measurement, 2002). We conceptualized the SAT-10 as a distal measure of general mathematics achievement.

Early Numeracy–Curriculum-Based Measurements (EN-CBM; Clarke & Shinn, 2004)

Based on principles of curriculum-based measurement (Shinn, 1989), we utilized four one-minute fluency assessments of key mathematical concepts that have been validated for use with kindergarten students (Chard et al., 2005; Clarke, Baker, Smolkowski, & Chard, 2008):

1. *Oral Counting (OC)*. The OC measure requires students to orally rote count as high as possible without making an error. Concurrent and predictive validity range from .46 to .72 with other published standardized measures.
2. *Number Identification Measure (NI)*. The NI measure requires students to orally identify numbers between 0 and 10 when presented with a set of printed number symbols. Concurrent and predictive validity range from .62 to .65 with other published standardized measures.
3. *Quantity Discrimination Measure (QD)*. The QD measure requires students to name which of two visually presented numbers between 0 and 10 is greater. Concurrent and predictive validities range from .64 to .72.

4. *Missing Number Measure (MN)*. The MN measure requires students to name the missing number from a string of numbers (0–10). Students are given strings of three numbers with the first, middle, or last number of the string missing. Concurrent and predictive validities range from .46 to .63. A total score, EN-CBM, was calculated by summing the raw scores from the four measures.

Test of Early Mathematics Ability–Third Edition (TEMA-3)

The TEMA-3 (Pro Ed, 2007) is a norm-referenced individually administered measure of early mathematics for children ages 3 to 8 years, 11 months focused on procedures and understanding of whole numbers. It measures both formal and informal mathematics, including skills related to counting, number facts and calculations, and related mathematical concepts. Test authors report alternate-form reliability of .97 and test-retest reliability ranges from .82 to .93. Concurrent validity with other criterion measures of mathematics is reported as ranging from .54 to .91.

Stanford Achievement Test–Tenth Edition (SAT-10)

The SAT-10 (Harcourt Brace Educational Measurement, 2002) is a group-administered, norm-referenced exam for kindergarten through grade 12 students and at kindergarten includes items related to number, geometry, and measurement. The *Math Problem Solving* subtest assesses problem solving and mathematical reasoning. The *Math Procedures* subtest assesses computational fluency. A standardized achievement test, the SAT-10 has adequate and well-reported validity ($r = .67$) and reliability ($r = .93$).

Fidelity of Implementation

During the study, the research team observed each teacher implementing mathematics lessons on three separate occasions. These observations were coordinated with the teacher and occurred in the fall, winter, and spring. The fidelity measure noted whether each ELM lesson activity occurred and the extent to which the lesson matched the intended delivery specified in the ELM curriculum. Each activity was rated as either “full implementation” (1.0), “partial implementation” (.5), or “not taught” (.0). Interrater reliability was obtained for 20% of all observations. Moderate to high interobserver reliability was found, with an ICC of .83. Results indicate that ELM teachers implemented ELM with fidelity across all three time points: fall ($M = .86$, $SD = .13$), winter ($M = .87$, $SD = .15$), and spring ($M = .87$, $SD = .14$).

Procedures

Data Collection

Trained staff administered all student measures and were required to obtain interrater reliability coefficients of .85 prior to collecting data with students. Follow-up trainings were conducted prior to posttesting to ensure continued reliable data collection. Oregon TEMA and CBM training consisted of a four-hour training session prior to pretest and posttest administration. Texas staff participated in four-hour TEMA trainings and a two-hour CBM training. All staff had to first meet the reliability standard during the in-training

reliability test and then had to meet the standard during an in-school shadow-coding session. Staff were trained to administer the SAT-10 during a four-hour training.

More than 11 hours of training were provided to staff completing lesson observations in both ELM and control classrooms. Trainings consisted of lecture, video-based practice, and video-based reliability checks as well as paired observations in schools and debriefing sessions. Staff met a reliability standard of .80 in and out of the classroom before independently conducting observations in the schools. Follow-up trainings were conducted prior to each data collection period to ensure continued reliable data collection. Maintenance reliability data were collected for at least 20% of all observations each round.

Professional Development

In the fall, before the study began, all ELM and control teachers met for two to four hours to discuss project details and logistics, learn how to submit instructional logs, and complete instructional and demographic surveys. In both states, ELM teachers also received three four-hour ELM curriculum trainings throughout the school year (fall, winter, and spring). In Oregon, the lead author of ELM provided the trainings. In Texas, the lead author led the first training and provided support to the Texas project coordinator who led subsequent trainings. Training addressed lesson content, study logistics, and classroom management. Training focused on teachers implementing the ELM curriculum as specified in the ELM teacher manual. Specific teacher behaviors were reviewed and practiced, including teacher models of key mathematic concepts, frequent guided practice opportunities (to the whole class and individual students), frequent independent practice opportunities (verbal and written, individually and with partners) and how to provide praise as well as positive and specific confirmations, monitor responses, and immediately correct student errors.

Protections Against Treatment Diffusion

To minimize treatment diffusion, control teachers did not participate in ELM curriculum trainings, and intervention teachers were asked not to share ELM ideas and materials with their non-ELM colleagues. Control teachers were asked to adhere to standard district practices and teach mathematics lessons in whichever fashion they usually did. Control teachers were formally observed conducting their math lessons three times throughout the year using the ELM fidelity of implementation observation instrument, which included a specific item addressing whether or not the teacher used the ELM curriculum and ELM materials. Fidelity observations and classroom visits by project staff revealed that treatment diffusion did not occur.

Statistical Analysis

We assessed intervention effects on each of the primary outcomes with a nested time by condition analysis (Murray, 1998) to account for students nested within classrooms. This analysis tests differences between conditions on change in outcomes from the beginning of kindergarten (T_1) to the end of kindergarten (T_2) clustered within classrooms. The nested time by condition analysis accounts for autocorrelation among assessments within individual students and the intraclass correlation associated with multiple students nested within the same classrooms. As a test of net differences, it also provides an unbiased and straightforward interpretation of the results (Cribbie & Jamieson, 2000; Fitzmaurice, Laird,

& Ware, 2004). The specific model tests time, coded 0 at T_1 and 1 at T_2 , condition, coded 0 for control and 1 for ELM. For the analyses of SAT-10 measures, we did not have a pretest assessment with the same measure. We analyzed these data with a mixed-model analysis of covariance (Murray) using the pretest TEMA raw score as the covariate. As a multilevel model, this approach also accounts for the intraclass correlation associated with multiple students nested within the same classrooms. With 129 classrooms, all tests used 126 degrees of freedom.

We also tested pretest achievement and state as moderators for each outcome measure. The effect for time by condition by pretest achievement estimates the difference in gains in the dependent variable at varying levels of achievement in the fall of kindergarten. State was coded with Oregon as 0 and Texas as 1.

Model Estimation

We fit models to our data with SAS PROC MIXED version 9.1 (SAS Institute, 2004) using restricted maximum likelihood (REML), generally recommended for multilevel models (Hox, 2002). From each model, we estimated fixed effects and variance components. This analysis approach included all data—whether or not a student's scores were present at both time points—to estimate differences between assessment times and between conditions. Maximum likelihood estimation for the time by condition analysis uses all available data to provide potentially unbiased results even in the face of substantial attrition, provided the missing data were missing at random (Schafer & Graham, 2002). In the present study, we did not believe that attrition or other missing data represented a meaningful departure from the missing at random assumption, meaning that missing data did not likely depend on unobserved determinants of the outcomes of interest (Little & Rubin, 2002).

The TEMA, for example, provides raw, scaled, and percentile scores. We presented results with raw scores, but the results of analyses using other TEMA scores (e.g., scaled scores) differed only slightly (e.g., p -values differed by less than .0015) and would not have changed the interpretation of any results.

Effect Sizes

To ease interpretation, we computed an effect size, Hedges' g (Hedges, 1981), for each fixed effect. Hedges' g , recommended by the What Works Clearinghouse (2011), represents an individual-level effect size comparable to Cohen's d (Cohen, 1988; Rosnow & Rosenthal, 2008).

RESULTS

Table 3 presents means, standard deviations, and sample sizes for each measure by assessment time and condition. We tested the sample for attrition effects, and differential attrition in particular. We addressed our research hypotheses with the nested time by condition model previously defined. Complete model results can be found in Tables 4 and 5 and include our primary outcomes, the TEMA, EN-CBM total, and SAT-10 tests.

Table 3. Descriptive statistics for mathematics measures by condition and assessment time

Measure	Statistic	ELM			Control		
		T ₁	T ₂	T ₃	T ₁	T ₂	T ₃
TEMA Raw Score	<i>M</i>	19.7	34.0		20.1	33.2	
	(<i>SD</i>)	(10.46)	(9.86)		(10.04)	(10.05)	
EN-CBM	<i>M</i>	76.1	164.0		76.7	158.4	
	(<i>SD</i>)	(52.17)	(50.86)		(51.44)	(51.34)	
SAT-10 Problem Solving	<i>M</i>			28.3			28.4
	(<i>SD</i>)			(7.99)			(8.23)
SAT-10 Procedures	<i>M</i>			17.51			17.44
	(<i>SD</i>)			(6.53)			(6.40)
Sample Size	<i>N</i>	1227	1134	893	1050	982	788

Note. The sample sizes represent the maximum students available across measures for each assessment period. Many of the students missing data at one assessment have data at other assessment times. At T₁, 55 students in the ELM condition and 51 students in the control condition did not have valid TEMA scores. At T₂ and T₃, all sample sizes were within four students of the maximum sample sizes listed in the table. Test of Early Mathematics Ability (TEMA), Early Numeracy–Curriculum-Based Measures (EN–CBM), Stanford Achievement Test Tenth Edition (SAT-10).

Attrition

Participant attrition, also called experimental mortality, can pose a threat to both the external and internal validity of a study (Barry, 2005; Shadish, Cook, & Campbell, 2002). Student attrition was defined as students with data at T₁ (pretest) but missing data at T₂ (posttest; TEMA or CBM) or T₃ (follow-up; SAT-10). For TEMA, we experienced 10.5% attrition at T₂, with 9.3% of the sample missing data in control classrooms and 11.5% missing in intervention classrooms ($\chi^2 = 3.39$, $df = 1$, $p = .0656$). EN-CBM scores were missing for 10.6% of students, with 9.3% of the students missing data in control classrooms and 11.6% missing in intervention classrooms ($\chi^2 = 3.82$, $df = 1$, $p = .0508$). Finally, we experienced 29.9% attrition at T₃ on SAT-10 data, with 27.5% of the sample missing data in control classrooms and 30.1% missing in intervention classrooms ($\chi^2 = 2.18$, $df = 1$, $p = .1396$). Attrition rates were slightly higher for intervention classrooms, but for each of the χ^2 tests, $p > .05$. Thus, we reject the hypothesis that attrition rates differed between conditions.

Although differential rates of attrition are undesirable, differential scores on math tests present a far greater threat to internal validity. We conducted an analysis to test whether student math scores were differentially affected by attrition across conditions. The analysis examined the effects of condition, attrition status, and their interaction on pretest scores of the TEMA, EN-CBM, and SAT-10 within a mixed-model ANOVA that nested students' T₁ scores within classrooms and condition (Murray, 1998) and predicts them with T₂ or T₃ attrition variables defined by the presence or absence of data at that time. We found no statistically significant interactions between attrition and condition at T₂ for the TEMA ($p = .6906$) or EN-CBM ($p = .2697$) or at T₃ for the TEMA ($p = .7656$) or EN-CBM ($p = .4855$). The attrition analysis demonstrates that while attrition was slightly more likely among intervention classrooms, those students lost to attrition at T₂ and T₃ did not differ by condition on their pretest TEMA or CBM scores.

Table 4. Fixed effect, variance component, and specific effect estimates for main effects of condition on mathematics outcomes

Effect or Statistic		TEMA Raw Score	EN-CBM	SAT-10 Problem Solving	SAT-10 Procedures
Fixed effects	Intercept	18.82*** (.72)	72.24*** (3.47)	17.95*** (.46)	9.80*** (.39)
	Time	13.22*** (.40)	81.06*** (2.60)		
	Condition ^a	.10 (.99)	1.37 (4.78)	-.07 (.49)	.11 (.39)
	Time × Condition	1.08~ (.55)	5.23 (3.58)		
Variance components	Pretest TEMA			.51*** (.01)	.38*** (.01)
	Residual	19.00*** (.63)	690.91*** (22.40)	26.23*** (.97)	23.04*** (.86)
	Student	58.35*** (2.07)	1448.16*** (55.26)		
	Classroom Intercept	23.34*** (3.64)	450.82*** (80.16)	5.21*** (.97)	2.77*** (.61)
	Classroom Gains	3.59*** (.62)	160.88*** (26.22)		
Hedges' <i>g</i>	Time × Condition	.108	.100	-.008	.016
<i>p</i> -value	Time × Condition	.0517	.1463	.8911	.7853
ICC		.159	.189	.166	.107

Note. Table entries show parameter estimates with standard errors in parentheses. Time is coded 0 for T1 and 1 for T2. Condition is coded 0 for control and 1 for ELM. All tests fixed effects used 126 *df*. Test of Early Mathematics Ability (TEMA), Early Numeracy–Curriculum-Based Measures (EN–CBM), Stanford Achievement Test Tenth Edition (SAT-10).

^aCondition effects for mixed-model analysis of covariance used to analyze SAT-10 problem solving and procedures measures. Pretest covariate was the TEMA pretest score.

p* < .05. *p* < .01. ****p* < .001. *****p* < .0001.

Table 5. Fixed effect, variance component, and specific effect estimates from a test-pretest TEMA as a moderator of condition effects on mathematics outcomes

	Effect or Statistic	TEMA Raw Score	EN-CBM	SAT-10 Problem Solving	SAT-10 Procedures
Fixed effects	Intercept	19.29*** (.26)	75.12*** (1.76)	27.66*** (.35)	16.92*** (.28)
	Time	13.38*** (.37)	82.02*** (2.40)		
	Condition	.00 (.36)	1.32 (2.41)	-.02 (.48)	.18 (.39)
	Time $\times$ Condition	1.00~ (.51)	4.15 (3.31)		
	TEMA ^a	1.00*** (.01)	4.36*** (.10)	.55*** (.02)	.40*** (.02)
	TEMA $\times$ Condition	.00 (.02)	-.10 (.14)	-.04 (.03)	-.03 (.03)
	TEMA $\times$ Time	-.22*** (.02)	-.84*** (.13)		
	TEMA $\times$ Time $\times$ Condition	-.08** (.03)	-.23 (.17)		
Variance components	Residual	.00 (.38)	-.77 (17.09)	26.76*** (.96)	23.22*** (.84)
	Student	3.19*** (.54)	133.83*** (22.63)		
	Classroom Intercept	.00 (.37)	255.20*** (20.72)	5.20*** (.96)	2.82*** (.61)
	Classroom Gains	15.50*** (.50)	641.55*** (20.36)		
<i>p</i> -value	TEMA $\times$ Condition			.9737	.6336
	TEMA $\times$ Time $\times$ Condition	.0039	.1887		

Note. Table entries show parameter estimates with standard errors in parentheses. Time is coded 0 for T1 and 1 for T2. Condition is coded 0 for control and 1 for ELM. Risk is coded 0 for no risk and 1 for students below the 40th percentile. All tests' fixed effects used 126 *df*. Test of Early Mathematics Ability (TEMA), Early Numeracy-Curriculum-Based Measures (EN-CBM), Stanford Achievement Test Tenth Edition (SAT-10).

^aThe moderator was the TEMA pretest value collected at the fall of kindergarten (*T*₁).

p* < .05. *p* < .01. ****p* < .001. *****p* < .0001.

Condition Effects

Although we expected the largest gains for students with some risk for math problems, we also anticipated that ELM would improve instruction for all students. We did not, however, find statistically significant condition effects for the TEMA, EN-CBM, or the SAT-10 (see Table 4). The condition differences for the TEMA raw scores were near statistical significance, $p = .0517$. The effect size for the full sample was $g = 0.11$.

Tests of Moderators

We tested two key moderators for this study. First, we tested state as a moderator, Oregon versus Texas. We did not expect the ELM curriculum to work differently by state, but it represents an important factor because (a) we recruited and randomly assigned classrooms within state and (b) states differed on a number of school practices (e.g., longer instructional period). State did not moderate condition effects for the primary outcomes: TEMA ($p = .5651$), EN-CBM ($p = .4470$), SAT-10 problem solving ($p = .4570$), or SAT-10 procedures ($p = .7718$). States also did not differ by initial values on the TEMA within the ELM ($p = .5721$) or control ($p = .2879$) conditions. Students in Texas schools, however, ended the year with higher scores within both the ELM classrooms ($p = .0310$) and control classrooms ($p = .0031$).

Dependence of Effects on Initial Achievement

We expected the ELM curriculum to offer a greater benefit to students who began the year with lower pretest scores than students who initially performed better on math assessments. We therefore tested pretest TEMA scores as a moderator of intervention effects. Initial achievement moderated condition effects for kindergarten TEMA scores ($p = .0039$) but not kindergarten EN-CBM ($p = .1887$) scores, first grade SAT-10 problem solving ($p = .9737$), and first grade SAT-10 procedures ($p = .6336$). Table 5 presents the results of this analysis.

Figure 2 shows the difference between conditions plotted against the TEMA normative percentile. The figure suggests that ELM was efficacious for students below the 25th percentile. In this study, 50% of the participating students began the year below the 25th percentile, and 75% scored below the 50th percentile.

DISCUSSION

This randomized controlled trial examined the efficacy of the core kindergarten mathematics program, Early Learning in Mathematics (ELM), which is a 120-lesson program with four content strands: (a) number operations, (b) geometry, (c) measurement, and (d) vocabulary. Overall, students in ELM classrooms achieved mathematic outcomes that were not significantly different than those achieved by students in control classrooms, that is, there was no overall impact found for the entire sample. However, students who began the school year with low initial mathematics skills (below the 25th percentile) had stronger outcomes on the TEMA but not the EN-CBM. Nor was there a significant effect of treatment on the first grade standardized assessments of mathematics. There were no differences in outcomes based on study location.

Limitations and Implications for Practice and Future Research

The findings from the study should be interpreted with an awareness of a number of critical limitations. Due to working across multiple districts, each with distinct parameters governing the release of student demographic data, we were unable to collect and report individual student demographic data. As with any study completed in specific states and with a specific sample, caution must be applied when interpreting results. Given the complexity of conducting social science research, increased attention has been paid to the importance of replication (Duncan, Claessens, Engel, & Dowsett, 2013), where the null hypothesis is exposed to repeated attempts at falsification with an array of geographically and demographically diverse samples.

A challenge when developing and evaluating a theory of change model is the selection of measures that map directly onto the constructs specified in the theory of change model. In this study, there is some concern about how directly we were able to evaluate performance on underlying constructs including the use of EN-CBM as a proximal assessment of procedural fluency and the TEMA as a measure of conceptual understanding. The TEMA, because of its focus on whole number understanding, was not able to assess gains in other areas of ELM (i.e., geometry and measurement). The EN-CBM measures, although assessing procedural fluency on key tasks such as identifying numbers and making magnitude comparisons, targeted skills more aligned with objectives covered earlier in the year. A measure more aligned with procedural fluency in basic addition and subtraction combinations would more directly map onto end-of-year objectives. In addition, one strength of CBM measures is the ability to monitor growth over time. In our study, the EN-CBM measures were used as pre-post measures of student achievement. Future work should consider using CBM measures at multiple time points to gain a better understanding of student growth, including the possibility of nonlinear growth on key constructs. We also created a total EN-CBM score

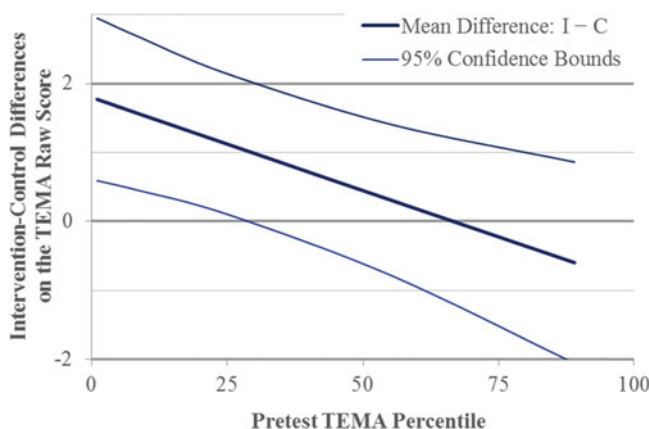


Figure 2. This figure shows differences between conditions (vertical axis) plotted as a function of pretest TEMA normative percentiles (horizontal axis). The darker line represents the mean difference between conditions, and the two lighter, outer lines show the 95% confidence bounds on difference. The figure shows that the difference between conditions becomes statistically significantly positive, implying a benefit from ELM, for students below the 25th percentile. For students above the 25th percentile, the confidence bounds include zero difference, which indicates no effect for this sample. *Note:* Test of Early Mathematics Ability (TEMA).

by summing the raw scores of four individual measures that varied in their overall range of scores (e.g., Oral Counting scores compared to missing Number scores), resulting in a total score that was more dependent on scores from measures that were easier for kindergarten students to master. Thus, the EN-CBM total score may not have been the best metric to detect differences between students. The use of the SAT-10 as the first grade outcome measure, although feasible due to being group administered, did not represent a strong outcome measure in terms of alignment to the core objectives of the ELM program as specified in the internal content dimension of the ELM theory of change. We have in subsequent studies utilized a researcher-developed proximal assessment that more directly links to the concepts taught in ELM and the application of those skills to fluently complete procedures.

In this article, although we examined whether or not ELM worked, we did not examine why ELM may have worked or not worked, in other words, what specific elements of the program, teacher delivery, or other variables may have mediated or moderated results. We have begun to explore various associations between critical variables and student outcomes to flesh out how exactly ELM might impact student achievement. In previous studies using the same data set as reported on in this article (Doabler et al., 2015; Doabler et al., 2014), we reported results from the use of an observational tool (Classroom Observations of Student-Teacher Interactions–Mathematics [COSTI-M]; Doabler et al., 2015) designed to document key elements of our theory of change model related to the quantity and quality of explicit teacher and student instructional interactions. The COSTI-M was used during observations in all participating kindergarten classrooms, regardless of condition, to measure the intensity of explicit instruction. Approximately 400 observations in 129 kindergarten classrooms were conducted. We were particularly interested in using the COSTI-M to examine the relationship between the rate of explicit instructional interactions and student mathematics achievement. Specifically, the COSTI-M captured three key components of instructional interactions hypothesized to potentially mediate student mathematics achievement: explicit teacher demonstrations, student practice opportunities, and timely academic feedback. Two key findings emerged from this work. First, teachers in ELM classrooms provided significantly higher rates of practice (Doabler et al., 2015) and second, students in classrooms with higher rates of practice opportunities made substantively important gains in critical mathematics outcome (Doabler et al., 2015). Thus, there seems to be emerging support for the ELM theory of change model and the hypothesis that critical teacher and student interactions, facilitated by the ELM curriculum, mediate mathematics achievement. It should be noted that although we theorize that the construction of ELM was the primary cause of changes in teacher behavior, treatment teachers were provided professional development on the ELM curriculum and on effective teaching strategies and behaviors. A stronger research design would have provided control teachers with professional development on the same teaching strategies and behaviors, thus eliminating professional development as an alternate possible cause and confound when interpreting the study results.

Other studies of core curricula have found evidence that programs designed to be explicit and systematic positively impact student achievement (Agodini & Harris, 2010; Clements, Agodini, & Harris, 2013). For example, Agodini and Harris (2010) investigated the effectiveness of four commercially available core elementary mathematics curricula and found that the mathematics achievement of students who were provided instruction with an explicit and systematic curriculum (i.e., *Math Expressions*) was significantly greater than the math achievement of students who were provided instruction with constructivist curriculums (i.e., *Scott Foresman* and *Investigations*) in first grade and that both *Math Expressions*

and *Saxon* (another explicit and systematic curriculum) produced greater achievement gains in second grade in comparison to *Scott Foresman*.

The findings from the present study provide a number of interesting considerations as we contextualize the study within the framework of early mathematics instruction. Although the results were not significant with the whole sample, there was some evidence that ELM did have a positive impact on students with lower skills at the beginning of the kindergarten year. Two potential implications and uses of ELM from the results should be considered. First, for schools and districts serving an overall at-risk population, it may be a judicious use of resources to implement a core program, such as ELM, that has evidence of effectiveness for the population of students being served by the school or district. This is of particular interest in kindergarten where the availability of research-based interventions is limited (Dyson et al., 2011) and the time available to implement them is also greatly constrained. So to that end, any impact that occurs within the context of core instruction and does not require additional instructional time or additional services could be considered promising. Second, for schools that do have the resources to provide intervention to at-risk kindergarten students, the use of ELM as a targeted intervention program may be a viable option to ensure that those students are provided needed instruction. As evidence of the importance of a successful early start in mathematics mounts (e.g., Morgan et al., 2009), various delivery options for ELM and similar curricula increase the potential for widespread use and impact.

Advances in the design and delivery of reading instruction can be used as a guide when considering implications for practice and future research for ELM specifically and early mathematics generally. Findings that growth on critical reading variables was influenced by initial skill and type of instruction provided (e.g., Connor, Morrison, & Katch, 2004; Foorman, Francis, Fletcher, Schatschneider, and Mehta, 1998), similar to the pattern of results on ELM, led to the development and testing of instruction specifically matched to student skill profiles resulting in greater gains (e.g., Al Otaiba, et al., 2011; Connor et al., 2011). Similar investigations in the early mathematics area are warranted and would potentially lead researchers and curriculum developers to a greater understanding of student instruction interactions and more nuanced ways of delivering targeted instruction.

It should be noted that even for studies that demonstrate a positive impact, the achievement gap is rarely fully bridged for students who do respond (Starkey & Klein, 2008) and that within studies that demonstrate a positive impact there are still students who don't respond (Fuchs, Fuchs, & Compton, 2012). Both of those trends are evident in this study. Thus, despite positive findings for specific curricula or interventions, there will be a need for more intensive services to either fully reduce the achievement gap or to meet the needs of students who do not make progress similar to their peers as well as a need for the development of and research on multitiered interventions. Lastly, given that many students enter kindergarten with an already-existing deficit in their mathematics skills that may be too severe to be addressed within the time frame of a single kindergarten year, consideration should be paid to examining ways to strengthen the transition from the informal mathematics knowledge students develop prior to school to the formal mathematics they receive in kindergarten. For example, such efforts have focused on working with at-risk students by providing an effective pre-kindergarten program with targeted professional development for kindergarten teachers (Sarama, Clements, Wolfe, & Spitler, 2012) or with an effective kindergarten curricula to provide at-risk students with two years of intervention across a significant transition period (Starkey, Klein, Baker, & Clarke, 2012). It is likely that researchers and schools will need to explore the potential of multiyear interventions if they are to meet the goal of bridging achievement gaps.

CONCLUSION

Despite the importance of a successful start in mathematics, there is limited research examining the efficacy of core curricula in kindergarten. Given that kindergarten often represents the first exposure to formal mathematics and the core curricula functions as the first and perhaps only exposure to mathematics for at-risk learners, continued research on the efficacy of core kindergarten curricula is warranted. The findings from the current study offer a limited but promising avenue for further exploration as the field attempts to ensure that all children learn foundational mathematics.

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